

Matrix Theory - Fall 2026

Practice Problem Set 07

1. For each map below, decide whether it is a linear transformation. If it is, find its transformation matrix T ; if it is not, show exactly which property (additivity or homogeneity) fails with a concrete counterexample.

(a) $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(\mathbf{x}) = \begin{bmatrix} x_1 + x_2 \\ 3x_1 \end{bmatrix}$

(b) $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(\mathbf{x}) = \begin{bmatrix} x_1 + 1 \\ x_2 \end{bmatrix}$ (a translation)

(c) $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(\mathbf{x}) = \begin{bmatrix} x_1 x_2 \\ x_2 \end{bmatrix}$

(d) $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(\mathbf{x}) = \begin{bmatrix} |x_1| \\ x_2 \end{bmatrix}$

(e) $f: \mathbb{R}^3 \rightarrow \mathbb{R}$, $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$ for a fixed $\mathbf{a} \in \mathbb{R}^3$

(f) (*Theory*) Show that every linear transformation satisfies $f(\mathbf{0}) = \mathbf{0}$.

2. (a) A linear map $f: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ satisfies $f(\mathbf{e}_1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $f(\mathbf{e}_2) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$, $f(\mathbf{e}_3) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Write down T and compute $f([2 \ -1 \ 1]^T)$.

- (b) A linear map $g: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is known only through its action on a *non-standard* basis:

$$g\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad g\left(\begin{bmatrix} 1 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Find the transformation matrix of g .

3. Rotations. Let $R(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ be the rotation matrix that rotates vectors counterclockwise by θ . Show that $\|R(\theta)\mathbf{x}\| = \|\mathbf{x}\|$ and $\langle R(\theta)\mathbf{x}, R(\theta)\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$ for all $\mathbf{x}, \mathbf{y}$. What does this say about lengths and angles under rotation?

4. Reflections. Let $\mathbf{e}_v$ be a unit vector and $F = 2\mathbf{e}_v\mathbf{e}_v^T - I$ the reflection about the line spanned by $\mathbf{e}_v$.

- (a) Show that $F^T = F$ and $F^2 = I$. What does this say about the reflection?

- (b) Compute F when $\mathbf{e}_v = [\cos \theta \ \sin \theta]^T$.

- (c) Let F_1, F_2 be reflections about lines at angles θ_1 and θ_2 with the x -axis. Show that $F_2 F_1 = R(2(\theta_2 - \theta_1))$.

5. Projections. Let $\mathbf{v} = [3 \ 4]^T$ and let P be the matrix that projects onto $\mathbf{v}$.

- (a) Compute the projection matrix P .

- (b) Show that $P^2 = P$ in general (not just for this $\mathbf{v}$), and explain geometrically why projecting twice changes nothing.
- (c) Find $\text{Col}(P)$ and $\text{Null}(P)$, and state the rank and nullity.
- (d) Show that $I - P$ is also a projection, and identify the line it projects onto.
- (e) For $\mathbf{y} = [3 \ 4]^T$, find the complete set $\{\mathbf{x} : P\mathbf{x} = \mathbf{y}\}$ and describe it geometrically.

6. Shear. Let $T_x(k) = \begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$.

- (a) Find the image of the unit square (vertices $\mathbf{0}, \mathbf{e}_1, \mathbf{e}_1 + \mathbf{e}_2, \mathbf{e}_2$). Using the $\mathbb{R}^2$ cross product $\mathbf{u} \times \mathbf{v} = u_1 v_2 - u_2 v_1$ from the previous problem set, show that the area of the image equals 1 for every k .
- (b) Show that $T_x(k)T_x(\ell) = T_x(k + \ell)$.
- (c) Which nonzero vectors keep their direction under $T_x(k)$ for $k \neq 0$? Is $T_x(k)$ a rotation for any $k \neq 0$?

7. Overview: In vector calculus, a differentiable map $\mathbf{g} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is, near a point $\mathbf{p}_0$, approximately linear:

$$\mathbf{g}(\mathbf{p}_0 + \mathbf{h}) \approx \mathbf{g}(\mathbf{p}_0) + J(\mathbf{p}_0) \mathbf{h}, \quad \text{where } J_{ij} = \left. \frac{\partial g_i}{\partial p_j} \right|_{\mathbf{p}_0}.$$

The Jacobian J can be considered as the transformation matrix of the *local* linear map $\mathbf{h} \mapsto J\mathbf{h}$. Its columns are $J\mathbf{e}_1$ and $J\mathbf{e}_2$: the images of tiny steps along each input coordinate.

Problem Statement: Consider a point p in two-dimensional space represented in polar coordinates as $\mathbf{p} = [r \ \theta]^T$. Let $\mathbf{g}(\mathbf{p})$ be the map from polar coordinates to Cartesian coordinates. The map can be expressed as $\mathbf{g}(\mathbf{p}) = \begin{bmatrix} r \cos \theta \\ r \sin \theta \end{bmatrix}$.

- (a) Compute the Jacobian $J(\mathbf{p}_0)$.
- (b) Show that $J = R(\theta) \begin{bmatrix} 1 & 0 \\ 0 & r \end{bmatrix}$. Describe in words the local action of $\mathbf{g}$ as “first _____, then _____”.
- (c) Interpret the two columns of J geometrically: in which directions do small changes in r and in θ move the point (x, y) , and by how much?
- (d) A tiny rectangle of sides $\Delta r, \Delta \theta$ in the (r, θ) -plane is mapped (approximately) to the parallelogram spanned by $J\mathbf{e}_1 \Delta r$ and $J\mathbf{e}_2 \Delta \theta$. Using the $\mathbb{R}^2$ cross product, show its area is $r \Delta r \Delta \theta$.

8. Overview: Recall from Set 06 that for $\boldsymbol{\omega} \in \mathbb{R}^3$, the skew-symmetric matrix $\hat{\boldsymbol{\omega}}$ satisfies $\hat{\boldsymbol{\omega}} \mathbf{r} = \boldsymbol{\omega} \times \mathbf{r}$, which is the velocity of the point $\mathbf{r}$ of a rigid body spinning with angular velocity $\boldsymbol{\omega}$. The four fundamental subspaces of $\hat{\boldsymbol{\omega}}$ have a clean mechanical meaning.

Problem Statement: Let $\boldsymbol{\omega} = [1 \ 2 \ 2]^T$.

- (a) Write $\hat{\boldsymbol{\omega}}$. Find $\text{Null}(\hat{\boldsymbol{\omega}})$. These are the points that do not move under the rotation, i.e., the axis of rotation.
- (b) Show that $\text{Col}(\hat{\boldsymbol{\omega}})$ is the plane $\boldsymbol{\omega}^\perp$. These are the directions of motion of points under the rotation, i.e., the plane perpendicular to the axis of rotation.

- (c) Suppose you observe a velocity $\mathbf{v}$ and want determine the position $\mathbf{r}$ such that $\boldsymbol{\omega} \times \mathbf{r} = \mathbf{v}$. Express this cross product as a matrix equation $\hat{\boldsymbol{\omega}} \mathbf{r} = \mathbf{v}$.

Explain why a solution exists *only if* $\mathbf{v} \perp \boldsymbol{\omega}$, and why it is never unique. For $\mathbf{v} = [2 \ -1 \ 0]^T$, verify that $\mathbf{r}_p = \frac{\mathbf{v} \times \boldsymbol{\omega}}{\|\boldsymbol{\omega}\|^2}$ is a solution and write the complete solution set. For $\mathbf{v} = [1 \ 0 \ 0]^T$, show there is no solution.

9. Gradient as a linear map. For a differentiable $f : \mathbb{R}^3 \rightarrow \mathbb{R}$, the directional derivative $D_{\mathbf{u}}f(\mathbf{p}) = \nabla f(\mathbf{p})^T \mathbf{u}$ is a linear transformation of $\mathbf{u}$ whose 1×3 transformation matrix is the row $\nabla f(\mathbf{p})^T$.

- (a) Let $f(x, y, z) = x^2 + y^2 + z^2$ and $\mathbf{p} = (1, 2, 2)$. Write the 1×3 matrix $J = \nabla f(\mathbf{p})^T$.
- (b) Find a basis for $\text{Null}(J)$. Explain why $\text{Null}(J)$ is exactly the set of directions in which f does not change to first order, i.e., the tangent plane to the sphere $f = 9$ at $\mathbf{p}$.
- (c) Using $\text{Row}(J) \perp \text{Null}(J)$, give a one-line linear-algebra proof of the vector-calculus fact that *the gradient is perpendicular to the level surface*.

10. Let

$$A = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 4 \\ 3 & 6 & 1 & 5 \end{bmatrix}.$$

- (a) Find the rank of A . Which columns are linear combinations of earlier columns, and with what coefficients?
- (b) Find a basis for each of $\text{Col}(A)$, $\text{Null}(A)$, $\text{Row}(A) = \text{Col}(A^T)$ and $\text{Null}(A^T)$.
- (c) Verify $\dim \text{Col}(A) + \dim \text{Null}(A) = 4$.
- (d) Verify that every null-space basis vector is orthogonal to every row, and that the left null-space vector is orthogonal to every column.

11. Let $A \in \mathbb{R}^{m \times n}$. Prove that every vector in $\text{Null}(A)$ is orthogonal to every vector in $\text{Row}(A)$.

12. **True or False.** Justify each answer in one or two sentences (a counterexample suffices for False).

- (a) The rank of a nonzero matrix is at least 1.
- (b) If A is 3×5 , then $A\mathbf{x} = \mathbf{0}$ has a nonzero solution.
- (c) If A is 5×3 , then $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b} \in \mathbb{R}^5$.
- (d) The columns of an $m \times n$ matrix can be linearly independent only if $m \geq n$.
- (e) If $\mathbf{x}_1$ and $\mathbf{x}_2$ both solve $A\mathbf{x} = \mathbf{b}$, then $\mathbf{x}_1 - \mathbf{x}_2 \in \text{Null}(A)$.
- (f) If $\mathbf{b} \neq \mathbf{0}$, the set of solutions of $A\mathbf{x} = \mathbf{b}$ is a subspace.
- (g) If $\text{Null}(A) = \{\mathbf{0}\}$, then $A\mathbf{x} = \mathbf{b}$ has exactly one solution for every $\mathbf{b}$.
- (h) $\text{Col}(AB) \subseteq \text{Col}(A)$ whenever AB is defined.
- (i) $\text{Null}(AB) \subseteq \text{Null}(A)$ whenever AB is defined and square.

13. Back and forward substitution.

- (a) Solve $U\mathbf{x} = \mathbf{b}$ by back substitution, where

$$U = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & -2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 10 \\ 6 \\ -4 \end{bmatrix}.$$

- (b) Count the exact number of multiplications, additions/subtractions and divisions used by back substitution on an $n \times n$ system.
- (c) Suppose the $(3, 3)$ entry of U in part (a) were 0 instead of -2 . For which b_3 does the system have a solution, and how many? Relate your answer to the rank and nullity of the modified U .

14. Elimination as premultiplication. Let

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 4 & 3 & 3 \\ 8 & 7 & 9 \end{bmatrix}.$$

- (a) Write the elementary matrices E_{21} , E_{31} , E_{32} that carry out Gaussian elimination (“subtract $\ell_{ij} \times$ row j from row i ”), and compute $U = E_{32}E_{31}E_{21}A$.
- (b) Solve $A\mathbf{x} = \mathbf{b}$ for $\mathbf{b} = [4 \ 10 \ 24]^T$ using $L\mathbf{y} = \mathbf{b}$ (forward) and then $U\mathbf{x} = \mathbf{y}$ (back).

15. Row operations vs. column operations. For $i \neq j$ define the elementary matrix $E_{ij}(k) = I + k\mathbf{e}_i\mathbf{e}_j^T \in \mathbb{R}^{n \times n}$.

- (a) Show that $E_{ij}(k)A$ is A with $k \times (\text{row } j)$ added to row i , while $A E_{ij}(k)$ is A with $k \times (\text{column } i)$ added to column j .
- (b) When is $E_{ij}(k)$ lower triangular, and when upper triangular?
- (c) Write the 3×3 elementary matrices that (i) swap rows 1 and 3, (ii) scale row 2 by $c \neq 0$. What happens when each is used as a postmultiplier instead?

16. Row and column operations together: completing the square. Let $A = \begin{bmatrix} 2 & 4 \\ 4 & 11 \end{bmatrix}$.

- (a) Find a lower triangular elementary E such that EA is upper triangular.
- (b) Now find an upper triangular elementary F such that $(EA)F$ is diagonal, D . Observe that $F = E^T$. Why is this expected when A is symmetric?

17. Show that the product of two lower triangular matrices is lower triangular, and that if both have unit diagonal, so does the product.

18. QR by Gram–Schmidt. Let

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

- (a) Apply Gram–Schmidt to the columns of A to obtain orthonormal $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$, and record $r_{ij} = \langle \mathbf{q}_i, \mathbf{a}_j \rangle$ ($i < j$) and r_{jj} (the length of the part of $\mathbf{a}_j$ orthogonal to earlier $\mathbf{q}$'s). Write $A = QR$.
- (b) Explain why R is upper triangular: what does the zero in position $(3, 1)$ say about how $\mathbf{a}_1$ is built from the $\mathbf{q}$'s?
- (c) Solve $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} = [2 \ 3 \ 3]^T$ by computing $Q^T\mathbf{b}$ and back-substituting in $R\mathbf{x} = Q^T\mathbf{b}$.
- (d) Show that if the columns of A are linearly dependent, Gram–Schmidt produces some $r_{jj} = 0$. What does this say about the rank of A ?

19. Geometric QR: rotations and reflections. Let $\mathbf{a} = [3 \ 4]^T$ and $A = \begin{bmatrix} 3 & 1 \\ 4 & 2 \end{bmatrix}$.

- (a) *Givens rotation.* Find the rotation $G = R(\theta)$ that maps $\mathbf{a}$ onto the positive x -axis, i.e. $G\mathbf{a} = [5 \ 0]^T$. Compute GA and write $A = QR$ with $Q = G^T$.
- (b) *Householder reflection.* Consider the mirror line along the bisector of $\mathbf{a}$ and $\mathbf{e}_1$, spanned by $\mathbf{w} = \mathbf{a}/\|\mathbf{a}\| + \mathbf{e}_1$. Explain geometrically why reflecting about this line sends $\mathbf{a}$ to $5\mathbf{e}_1$. Using $F = 2\mathbf{e}_w\mathbf{e}_w^T - I$, compute F and FA .
- (c) Show that the same F can be written as $H = I - 2\mathbf{u}\mathbf{u}^T$ with $\mathbf{u} \propto \mathbf{a} - \|\mathbf{a}\|\mathbf{e}_1$ (the standard Householder form). How is the line spanned by $\mathbf{u}$ related to the mirror line?